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# Depiction of library use in video games: a content analysis

Depiction of  
library use in  
video games

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## Abstract

**Purpose** – Researchers in information studies have examined fictional depictions of libraries in various mediums because these images can reflect and influence real-life experiences and attitudes. Video games, despite being relatively overlooked, are increasingly culturally relevant and can indicate library users' real needs and desires. This study investigates the ways in which video games depict characters using libraries to seek and use information.

**Design/methodology/approach** – A qualitative content analysis approach incorporating methods from information studies and game studies was applied. Tancheva's (2005) semiotic analysis of fictional libraries and Carr's (2019) textual approach provided the framing for the unique aspects of video games and their meanings. Carroll (2021)'s character analysis and Chatman (1996)'s theory on insiders–outsiders dynamic underpinned the data collection and analysis. The purposive sample included 15 video games released since 2010.

**Findings** – Video games depict game characters visiting libraries to solve short-term problems, to gain knowledge to improve themselves or to bond with others. Protagonists are often depicted as adventurers or outsiders who must adapt to unfamiliar places and situations to achieve their wider objectives. In these games, libraries provide useful documents, spaces or helpful guides and intellectuals who assist the protagonists. As outsiders, the protagonists seek information in libraries to help them learn about their environments and to immerse themselves in the local histories and cultures in their worlds. Overall, these depictions highlight both short- and long-term benefits of library use.

**Originality/value** – As with existing studies, the ways in which fictional library use appear in video games can suggest real needs and desires among library users. The findings from this study emphasise the importance of library services and spaces that help users both address short-term problems and immerse themselves in local concerns, with longer-term goals. Applying different research methods or lenses to analysing video games could deepen our understanding of what library users think and feel when they seek and use information in libraries.

**Keywords** Depictions of library use, Library use in video games, Fictional library use, Content analysis of video games, Information behaviour

**Paper type** Article

## 1. Introduction and motivation for study

Depictions of libraries and librarians in media such as films, literature and television are evident in information studies literature (Duxfield and Liew, 2022; Seale, 2008; Tewell, 2014). Less is known about the depictions of library use, and scholarly research on this is especially scant in the context of video games. This research addresses this gap. It was guided by the following questions:

- (1) Why are fictional libraries used in video games?
- (2) How are fictional library users depicted in terms of character traits and their information behaviour in video games?
- (3) How are the fictional characters' use of libraries depicted in video games?



In this study, library use refers to an individual's interactions with library spaces or services to support their personally defined processes or needs. Processes are how "*an individual uses a library or information to achieve an end*" (Fleming-May, 2011, p. 309). Information behaviour refers to the "*overarching term that includes intentional and planned behaviours (such as active seeking) as well as unintentional or serendipitous actions (such as glimpsing or encountering information), purposive behaviours that do not involve seeking such as rejecting or avoiding information, and the thoughts and emotions that accompany all of these. The term also encompasses the influence of contextual elements like situation, time, affect, culture, and geography*" (Case *et al.*, 2016, p. 370).

Video games are gaining cultural significance in many contemporary societies (Muriel and Crawford, 2018). The depiction of libraries and its function can to an extent, reflect or affect perceptions of the value of libraries (Hicks and Whippey, 2013) and, the expectation of the place and role of libraries in contemporary societies (Duxfield and Liew, 2022). With the supposition that fictional depictions of library use reflect some real experiences, expectations and attitudes among library users, such insights can be used to innovate, plan and design libraries spaces and information services accordingly. It is also anticipated that the insights from this study will encourage further scholarly research that examine the depictions of information behaviour around the use of libraries. This aligns with the call for researchers and libraries to prioritise users' perspectives in delivering services and measuring success and impact (Sputore and Fitzgibbons, 2017; Fleming-May, 2011).

## 2. Literature review

This research builds upon studies of fictional librarians, libraries and library use in popular culture. It also intersects with other strands in the fields of information studies and game studies.

### 2.1 Depiction of library use in popular media

*2.1.1 Librarians dominate the picture.* Fictional depictions of librarians and libraries are popular topics in information studies (Seale, 2008). Researchers often analyse positive or negative representations of librarians in film, literature and television. DeCandido (1999) and Seale (2008) for example, discuss whether the television show "Buffy the Vampire Slayer" subverts or perpetuates negative stereotypes of librarians. Some researchers regard the librarian in the show as a suave heroic character, but others argue that he is a stuffy Luddite (Hicks and Whippey, 2013). While different frameworks and methods account for this, these researchers also evaluate fictional librarians according to how the researchers want societies to perceive the library profession.

While the impact of negative stereotypes is unclear, these studies are rooted in concerns about cultural perceptions of libraries and librarians (Vassilakaki and Moniarou-Papaconstantinou, 2014). Many of these studies assume that fictional depictions reflect and influence cultural attitudes towards real-life libraries (Tewell, 2014). This aligns with social constructionism, an epistemological approach to studying social knowledge as a product of shared contexts (Drisko and Maschi, 2016). Some argue that these stereotypes affect the funding, use and perception of libraries in their societies (Rubin, 2015). However, not all depictions have been found to be negative. For instance, Duxfield and Liew (2022) uncover positive representations in science fiction novels. Understanding these positive and negative trends are important for libraries in contexts of economic pressure and ubiquitous access to online information (Rubin, 2015).

Few studies have focused on depictions of library users. If they recount the experiences of fictional users in libraries, it is to reflect on perceptions of librarians rather than on analysing

users' needs or behaviour. For example, [Radford and Radford \(2001\)](#) argue that fictional library users feel humiliated because librarians punish them for disrupting the order of their libraries. More positively, [Estill \(2007\)](#) and [Jarvis \(2008\)](#) cite examples of libraries as collaborative, conversational or romantic sanctuaries for all users, small groups, or "nomads". "Buffy" provides an example of the librarian and the protagonist collaborating as the "mind" and "body" respectively ([Estill, 2007](#)). While these are interesting points to consider, the researchers often take for granted the reasons why fictional users interact with libraries, and how this affects their experience.

*2.1.2 Information behaviour contextualises library use.* Researchers focusing on fictional library users' needs and desires often apply theories of information behaviour to their analysis. For example, [Hicks and Whippley \(2013\)](#) analyse "Buffy" and conclude that characters actively or indirectly seek information in a piecemeal rather than systematic manner. This includes tasks as disparate as completing homework or slaying a vampire. [Tewell \(2014\)](#) analyses how users feel incompetent and anxious in two television shows using the theory of library anxiety. [Pierce \(2004\)](#) compares theories of youth information seeking to fictional young people. They argue that fictional users are more self-determined than real-life young people, who often accept imposed queries. However, accordingly, both fictional and real young people develop their identities and connect with their wider communities through information seeking. [Urban \(2021\)](#) demonstrates that researchers can analyse video games through theories of information behaviour. Urban's focus is nevertheless, on players' interactions with fictional documents rather than depictions of library use.

## *2.2 Depiction of library use in video games*

A review of literature finds three Master's theses with a focus on the depictions of librarians and libraries in video games – [Holmström \(2019\)](#), [Kussler \(2020\)](#) and [Carroll \(2021\)](#). These studies highlight the unique aspects of these depictions and challenges in studying video games. While these scholarly investigations discuss the use of libraries, their primary focus is again on the libraries themselves.

*2.2.1 Game-specific variables affect depictions.* It has been found that similar to fictional libraries in other media, libraries in video games can help users learn, build relationships and connect to their communities ([Holmström, 2019](#); [Kussler, 2020](#)). They can also be intimidating, mysterious and lonely places for users, however. Aspects of video games as a medium modify these typical features. Games can express themes through game mechanics. For example, fictional users have been depicted as capable of improving their skills by reading books in libraries ([Kussler, 2020](#)). In-game book reading has been shown to help players understand the histories of fictional worlds, which leads to better decision-making ([Kussler, 2020](#)). Challenges are built into games for players to overcome. For example, intimidating librarians, hostile enemies or excessive rules could be negative for users in films, but in a video game context, these present lucid and at times, entertaining obstacles for game players ([Holmström, 2019](#); [Carroll, 2021](#)). These aspects are important to acknowledge in comparing video games to other media.

*2.2.2 Game genre affects depictions.* [Holmström \(2019\)](#) argues that the genre of video games affects the depiction of the appearance and function of librarians and libraries. [Tancheva's \(2005\)](#) use of semiotic theory to analyse the narrative context of depictions was drawn on. Tancheva argues that researchers must understand fictional libraries as fluid signs that shift according to genre, setting and subject. For example, a horror film enhances frightening aspects of a fictional library. [Holmström \(2019\)](#) finds this applies to the study of video games. Strategy games often depict libraries as small abstractions that generate resources, with only their exteriors visible. Other genres such as roleplaying games allow players to physically traverse and observe the interiors of libraries. Narrative context influences the depictions.

2.2.3 *Video game characters are multi-faceted.* Carroll (2021) adapts approaches from information studies and game studies to analyse librarian stereotypes in video games. They combine known librarian stereotypes (Seale, 2008) with different dimensions of game characters such as narrative importance, demographics and mechanical characteristics (Downs and Smith, 2009). Carroll (2021) codes narrative importance according to:

- (1) Primary characters – player controlled
- (2) Secondary characters – have direct influence/impact on a primary character
- (3) Tertiary characters – background or cosmetic

Carroll (2021)'s examination of demographic and mechanical characteristics is consistent with other observations in information studies and game studies. Librarian stereotypes are often gendered (Seale, 2008) and some game studies researchers such as Brock (2011) criticise video games for presenting heroes as straight white males exploring “exotic” places and defeating foreign villains. However, it should be noted that some games allow players to customise their character's appearance, species and gender. Demographics are widely noted in other media, but mechanics also influence character traits in video games (Kussler, 2020). Mechanics could include quest-giving, selling items or magical abilities. These characteristics determine how players can interact with the fictional worlds, other characters and their settings. Brock (2011) suggests that combat-oriented mechanics in games limit players' intellectual and emotional interactions. For example, players can read books in fictional libraries only if game designers provide the mechanics for them to do so, which affects the depictions.

### 2.3 Contextualising “library use”

2.3.1 *Users help define library use.* Fleming-May (2011) argues that information studies researchers define “library use” contextually, according to their research interests and identifies four categories of definitions in literature:

- (1) Use of the library as an abstraction, or general idea
- (2) Use of the library as an implement, or tool
- (3) Use of the library as a transaction or occurring within a discrete instance
- (4) Use of the library as a complex process (Fleming-May, 2011, p. 306)

Fleming-May argues for the “process” definition of library use. According to this, a user sets themselves tasks (such as finding a book) to fulfil personal processes or ends (such as completing a degree). Employing this definition emphasises the ends to which people visit libraries to satisfy their needs. Accordingly, this helps researchers and librarians understand how individuals might evaluate their own library use against their own needs and desires.

2.3.2 *Information behaviour emphasises people and contexts.* A concept closely related to library use is information behaviour (around the use of library). Case *et al.* (2016) define information behaviour as an umbrella term for a range of concepts and activities. They emphasise that it includes both individuals and groups' behaviour and the subjective reasons and contexts informing their actions. This is rooted in an understanding of behaviour as a variable process of distinct components (Vakkari, 2008). Commonly discussed components include:

- (1) Information need: a recognition “that your knowledge is inadequate to satisfy a goal that you have” (Case *et al.*, 2016, p. 6)
- (2) Information seeking: “a conscious effort to acquire information in response to a need or gap in our knowledge” (Case *et al.*, 2016, p. 6)

- (3) Information use: “what you do with the information acquired through seeking, serendipity, or other means. This includes applying – or ignoring – information to suit your goals or personal context” (Case *et al.*, 2016, p. 6)
- (4) Sharing (of information): “sharing information among a group of individuals, whether they are in contact through work or strictly socially. If the group of individuals share a common goal, then their sharing of information may be called collaboration” (Case *et al.*, 2016, pp. 374–375)

While these components commonly appear in everyday life, human information behaviour is often multifaceted, recursive and haphazard (Case *et al.*, 2016). It is also shaped by the social contexts and shared norms of individuals and groups (Floegel and Costello, 2021). Theories of information behaviour help researchers compare individuals and groups to understand how they act and why they might do so. Chatman’s (1996) theory of information poverty draws on sociological theories of insider and outsider dynamics to analyse why marginalised groups choose to avoid useful information sources and sharing with those outside their groups. Chatman defines “insiders” as people sharing common cultures and perspectives, while they define “outsiders” as those living marginalised lives, such as prisoners or rest-home residents. Accordingly, power imbalances and social norms prevent information flowing between these two worlds. Insiders share information amongst themselves, and outsiders avoid or control information to protect themselves. Other researchers have contested these conceptions. Floegel and Costello (2021) argue that individuals feel both inside and outside of groups in different situations, and that marginalised groups have rich rather than poor, information worlds. This is relevant in the context of studying the depiction of library use in that researchers often perceive fictional users as outsiders in libraries, albeit without specific reference to Chatman’s theories (Radford and Radford, 2001).

## 2.4 Scholarly examination of video games

*2.4.1 Researchers select elements of video games to analyse.* Game studies is a multidisciplinary field, focusing on understanding games in various contexts (Mäyrä, 2008). Video games include many elements that produce meaning, such as player interaction, visual elements or how underlying rules affect play (Bergonse, 2017). Researchers have debated which elements to prioritise in understanding games (Malliet, 2007). Some emphasise game mechanics (such as rules), while others emphasise the narrative elements (such as dialogue). Contemporary studies often mix these two approaches (Schmierbach, 2009).

Carr’s (2019) method of textual analysis is an example of a mixed approach. They aim to capture plural and ambiguous meanings in video games and reflect on them using theories (such as feminism). Their approach is to “fragment” games by identifying relevant elements, such as in-game objects (such as a book) or locations (such as a library), and iteratively considering the fragments through three overlapping lenses:

- (1) Structural – paying attention to relationships between elements of game design
- (2) Thematic – examining the play experience and signification of meaning
- (3) Intertextual – examining players’ subjective associations with other games and their personal context

For instance, a researcher could fragment an in-game book and analyse how the object improves a character’s abilities (structural lens), discusses gender roles (thematic lens), parodies a real-life book (intertextual lens) and how these lenses overlap. This approach examines how video games produce meaning and allows researchers to apply theories (such as information poverty) to delve beyond conventional readings of games.

*2.4.2 Researchers play games to understand them.* Researchers such as Carr (2019) argue that playing video games is essential for analysing them because they are both interactive and variable. Players control how they interact with a game, to an extent. It is therefore important to acknowledge that different players (or even play sessions) can produce different meanings in the same game (Malliet, 2007). This presents a challenge for researchers because individuals often prefer a certain playstyle. For example, players can focus on individual exploration, or on socialising with others. In response to this, it is recommended that researchers play a game in different styles to uncover and capture a broader range of possible meanings (Muriel and Crawford, 2018).

### *2.5 How does the literature influence this research?*

Previous studies have primarily focused on depictions of librarians and libraries in popular culture, not of library use. However, these studies indicate useful approaches to analysing depictions of library use. This study incorporates and builds upon concepts and findings from the literature, including the definition of library use as a “process” (Fleming-May, 2011). The methods and theories in information studies and game studies, including Carr (2019)’s textual analysis, Tancheva (2005)’s semiotic approach, Carroll (2021)’s character analysis and Chatman (1996)’s theory on insiders–outsiders dynamic underpin the data collection and analysis.

## **3. Research design**

Social constructionism underpins this study. A key premise is that knowledge is relative and contextual according to different social groups and cultures (Drisko and Maschi, 2016). In the context of this study, qualitative content analysis emphasises the subjective interpretation of researchers (Carr, 2019; Tancheva, 2005). In this study, the interpretation was that of the first-named author who played the video games as part of this study. It is acknowledged that the same video games are subjected to other possible interpretations by different players.

### *3.1 Sampling and assumption*

The target population was video games depicting library use. Kussler (2020) identified eighty-six video games depicting libraries and acknowledged more could exist. Holmström (2019) explained how different genres depict libraries in different ways. Noting this, in this research, video games were sampled across a range of genres. Kussler’s (2020) list of games was reviewed, and an Internet search was conducted applying variations of keywords, including “library”, “video games” and “fandom”. Publicly available YouTube videos featuring footage of the games were also reviewed to assess the depth and breadth of their depictions of library use (Carroll, 2021).

A final sample consisting of 15 video games with considerable depictions of library use were selected (Table 1). The sample included games released since 2010 and those available on Microsoft Windows, Nintendo Switch and PlayStation 4 (platforms accessible to the researchers). The sampling plan was unrestricted in terms of who created the video games and where they originated. It included big-budget Japanese games and low-budget “indie” games from Sweden and Ukraine.

### *3.2 Data collection*

The data collection and analysis followed Krippendorff’s (2018) components of content analysis: Unitising, sampling, recording/coding, reducing data, inferring context and narrating. Unitising involves choosing a unit of analysis. In the context of content analysis of video games, this is a piece of content that can be coded for analysis (Schmierbach, 2009).

| Title                                                                         | Acronym | Year | Platform          | Mechanical genre             | Thematic genre                                     | Country of origin                   | Time played |
|-------------------------------------------------------------------------------|---------|------|-------------------|------------------------------|----------------------------------------------------|-------------------------------------|-------------|
| Assassin's Creed Origins                                                      | ACOR    | 2017 | Microsoft Windows | Action-adventure, action-RPG | Historical                                         | Canada, France                      | 4 h18 m     |
| Dragon Age: Inquisition: Game of the Year Edition                             | DAIN    | 2014 | Microsoft Windows | Action-RPG                   | Fantasy, peri apocalyptic                          | Canada, United States of America    | 10 h45 m    |
| Dragon Quest XI S: Echoes of an Elusive Age - Definitive Edition              | DQ11    | 2019 | Nintendo Switch   | RPG                          | Fantasy, peri apocalyptic                          | Japan                               | 18 h22 m    |
| Eastshade                                                                     | EASH    | 2019 | Microsoft Windows | Adventure, photography       | Fantasy                                            | United States of America            | 4 h20 m     |
| Final Fantasy VIII Remastered                                                 | FFA8    | 2019 | Microsoft Windows | RPG                          | Science fiction, fantasy, peri apocalyptic         | Japan                               | 6 h36 m     |
| Fran Bow                                                                      | FRAB    | 2015 | Microsoft Windows | Adventure                    | Horror, fantasy                                    | Sweden                              | 3 h42 m     |
| Monkey Island 2 Special Edition: LeChuck's Revenge                            | MIS2    | 2010 | Microsoft Windows | Adventure                    | Comedy, pirate                                     | United States of America            | 3 h38 m     |
| NieR Replicant ver.1.22474487139 ...                                          | NIER    | 2021 | Microsoft Windows | Action-RPG                   | Post-apocalyptic, fantasy                          | Japan                               | 5 h21 m     |
| Night in the Woods: Weird Autumm Edition                                      | NTW     | 2017 | Microsoft Windows | Adventure                    | Fantasy, mystery, peri apocalyptic                 | United States of America, Australia | 4 h36 m     |
| OneShot                                                                       | ONSH    | 2016 | Microsoft Windows | Adventure                    | Post-modern, science fiction, peri apocalyptic     | United States of America, Japan     | 5 h37 m     |
| Persona 5                                                                     | PSA5    | 2017 | PlayStation 4     | RPG, simulation              | Fantasy, mystery, peri apocalyptic                 | Japan, Austria                      | 9 h24 m     |
| Psychonauts 2                                                                 | PSY2    | 2021 | Microsoft Windows | Platform game                | Comedy, science fiction, mystery, peri apocalyptic | United States of America            | 9 h36 m     |
| The Elder Scrolls V: Skyrim Special Edition                                   | SKIM    | 2016 | Microsoft Windows | Action-RPG                   | Fantasy, peri apocalyptic                          | United States of America            | 6 h48 m     |
| Stardew Valley                                                                | STVA    | 2016 | Microsoft Windows | Simulation, RPG              | Farming, fantasy                                   | United States of America            | 3 h36 m     |
| The Sinking City                                                              | TSIC    | 2019 | Microsoft Windows | Action-adventure             | Horror, mystery, peri apocalyptic                  | Ukraine, France                     | 5 h54 m     |
| <b>Source(s):</b> Table courtesy of Taylor and Liew (Authors of this article) |         |      |                   |                              |                                                    |                                     |             |

Depiction of library use in video games

**Table 1.**  
Video games in the study sample



In this study, the unit of analysis was any element relating to library use and users' information behaviour around the use of the library in the game. This included elements such as visual design, dialogue or cutscenes. Capturing this behaviour required collecting data from the stages before, during and after the user visited a library. The first-named author played the video games in the study sample from the beginning to the point when library use appeared. The library use segments were then repeated – using varying styles/approaches to play these sections. For example, one style involved following the game's narrative linearly while another involved experimenting with “odd” behaviours and choices (Muriel and Crawford, 2018). Testing these styles depended on what each game allowed players to do. Playing from the beginning helped the researcher immerse in the game setting and contextualised the depictions of library use within their narratives (Carr, 2019). Notes were recorded during and after each play session.

### 3.3 Data analysis

Data analysis involved coding and categorising data to identify key themes across the sampled video games (Drisko and Maschi, 2016). Each unit of analysis was considered to hold different layers of meaning, such as structural or intertextual (Carr, 2019). The narrative context, such as genre or setting (Tancheva, 2005) was noted and recorded. Repeated play sessions enabled these codes and categories to be iterated and refined (Drisko and Maschi, 2016). Themes and patterns across the study sample were identified. Carroll's (2021) character analysis informed the data analysis. Carroll analyses fictional librarians according to their importance, character traits, demographics and mechanical characteristics. This research analysed data using these different dimensions of character traits for the fictional library users.

## 4. Findings

The study sample consisted of single-player video games. They were primarily of the action, roleplaying game and adventure genres with some exceptions and stylistic variations (such as simulation or platform games). The remastered and remade versions of Final Fantasy VIII and Monkey Island 2, games originally released in the 1990s, were selected because of their popular status. Before each game was played, the researcher-player (first-named author) viewed online footage of the games to plan an approach to reaching scenes involving library use. The length of play varied as it depended on the size of the games and the stage of the stories in which a library appeared. A representative slice of each depiction of library use was obtained. After reaching the segments involving libraries, the play scenes were repeated in different playstyles, such as story-based, “weird”, hostile and roleplaying. An example of the “weird” style included attempting odd camera angles or testing a game's limits. Online footage of each game's ending was reviewed to situate the depictions of library use within their overall narrative contexts. This resulted in a rich collection of observations, notes and screenshots of the units of analysis.

### 4.1 The user experience

Elements related to library use were coded around the categories of library use, information behaviour and character traits.

4.1.1 *Who were the users as characters?* Table 2 summarises the characteristics of the protagonists in the games.

Of the 15 protagonists, nine were young, eight were male and nine were either Caucasian or Japanese. Four protagonists were potentially sexually queer. Aspects of these games (such as genre) at times posed challenges. Fantasy and science fiction games presented characters with different cultures and physical traits to humans. For example, Niko in ONSH was an

| Video games  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|--------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
|              | ACOR | DAIN | DQ11 | EASH | FFA8 | FRAB | MIS2 | NIER | NITW | ONSH | PSA5 | PSY2 | SKIM | STVA | TSIC |
| Age          |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| Young        |      |      | x    |      | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    |
| Adult        | x    | x    |      | x    |      |      |      |      |      |      |      |      |      |      |      |
| Ambiguous    |      |      |      | x    |      |      |      |      |      |      |      |      |      |      |      |
| Gender       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| Male         | x    |      | x    |      | x    |      | x    | x    | x    |      | x    | x    |      |      | x    |
| Female       |      |      |      |      |      | x    |      |      |      |      |      |      | x    |      |      |
| Optional     |      | x    |      |      |      |      |      |      |      | x    |      |      |      |      |      |
| Ambiguous    |      |      |      | x    |      |      |      |      |      |      |      |      |      |      |      |
| Sexuality    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| Heterosexual | x    |      |      |      | x    |      | x    |      |      |      | x    | x    |      |      |      |
| Queer        |      |      |      |      |      |      |      |      | x    |      |      |      |      |      |      |
| Optional     |      | x    |      |      |      |      |      |      |      |      |      |      | x    |      |      |
| Ambiguous    |      |      | x    | x    |      | x    |      |      |      | x    |      |      |      |      | x    |
| Ethnicity    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| Caucasian    |      |      |      |      |      | x    |      |      | x    |      |      | x    |      |      | x    |
| Japanese     |      |      |      |      | x    |      |      |      |      |      | x    |      |      |      |      |
| Egyptian     | x    |      | x    |      |      |      |      |      |      |      |      |      |      |      |      |
| Optional     |      | x    |      |      |      |      |      |      |      |      |      |      |      | x    |      |
| Ambiguous    |      |      |      | x    |      |      |      |      |      | x    |      |      | x    |      |      |

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Table 2.  
Demographic  
characteristics of video  
game protagonists

Depiction of  
library use in  
video games

anthropomorphic cat exploring a strange world of humans and robots. Games with optional demographics allowed players to create their own protagonists, allowing for variety in genders, species and other characteristics. At times, the protagonists, such as in EASH, were invisible to the player.

All the protagonists were characterised as adventurous and curious, exploring their worlds and interacting with others to achieve their goals. Visible appearances reflected their personalities or occupations. For example, Mae in NITW was an unemployed university dropout and wore casual clothing, while Charles in TSIC was a detective exploring an inhospitable town, and wore heavy clothing and carried tools such as spades and firearms. Games such as DAIN and SKIM allowed players to modify the protagonists' clothes and general appearance. The protagonists' mechanics included movement (such as running or jumping), interactions (such as reading or picking up objects) and fighting in the action and roleplaying games. These mechanics aligned with their personalities. The protagonists were more partial to physical or adventurous activities than intellectual activities. An example of a character is Guybrush Threepwood in MIS2. He is a Caucasian nineteen-year-old pirate with long hair and pirate garb. He is at times crafty and arrogant. At other times earnest, foolish and inexperienced. He talks with others, pick up objects (such as a lens), study them and use them to solve problems.

*4.1.2 How did the games begin?.* The protagonists, except in FFA8 and NIER, often appeared as outsiders in their worlds. They could be travellers (EASH), returning home (NITW) or incarcerated (SKIM). They had to learn to navigate unfamiliar and potentially hostile worlds. These worlds varied in their realism or fantasy. Some were based on historical places and periods, such as ancient Egypt (ACOR) or 1920s Massachusetts (TSIC). Others featured entirely fictional worlds with distinct regions and cultures (such as Erdrea in DQ11). However, all of them contained fantastical elements such as magic or non-human species. There were also highly literate worlds, with books or scrolls appearing in inns, workplaces and residences. This meant the protagonists could often pick up or read books outside of libraries. The high proportion of peri and post-apocalyptic games reflected how these worlds were often changing or decaying because of a mystery or political issue, such as a secretive cult (NITW).

The protagonists' goals revolved around a mix of personal and political issues. Their outsider status meant they sought to understand themselves in their settings. Personal and political issues intertwined. Roleplaying games allowed players to set their own objectives or eschewed a linear narrative. An example of an outsider is Niko in ONSH. Niko wakes up in a strange, post-apocalyptic world. They ask around to find a way to return to their world and mother. A robot explains that Niko is a messiah and must save the strange world. Niko believes saving this world may help them journey back to their mother.

*4.1.3 Why did users go to libraries?.* The protagonists used libraries to either solve short-term problems or to incrementally improve themselves and their relationships with others. Most were problem solvers, with four being improvement seekers (DAIN, FFA8, PSA5 and STVA).

Problem solvers had specific information needs that arose because of barriers to progressing in their journeys. They required knowledge about the world/setting, unique artefacts/objects or the help of others to overcome them. An example could be a difficult mystery that required historical knowledge to understand (NITW). For problem solvers, short-term needs led to library use rather than consistent and recurring needs. Improvement seekers visited libraries if the player chose to do so. Opportunities to visit libraries arose from discovering a mysterious object (STVA), being encouraged by another character (PSA5) or seeing the library on a map (DAIN). Improvement seekers brought artefacts/objects to bolster their abilities and relationships with people, often companions, in the libraries. This could be repeated throughout the stories.

Both types of protagonists encountered other characters or in-game messages directing them to visit libraries. This could be through information such as dialogue, item descriptions or maps. Some protagonists were depicted as being sceptical of libraries, perceiving them as

outdated or inefficient (FFA8 and NITW). Others were eager because of their familiarity or belief that libraries could help (FRAB, NIER and ONSH). The remaining protagonists appeared neutral, reacting little to the need to visit a library.

Protagonists physically travelled to the library spaces, except for in NITW where the game transitioned to the library, or in PSY2 where a character forced the protagonist into the library. Only in one game, FFA8, could a character access library information on a computer outside of its physical space. Visiting the libraries sometimes involved befriending wider communities to access their general area (such as a city) or the libraries themselves through passcodes or permissions (EASH, FRAB, MIS2, ONSH and SKIM). The others were freely accessible or appeared at certain stages of the game play.

An example of a problem solver is DQ11's protagonist, who searches for magical stones with his companions. Upon reaching a city containing a stone, they find that a witch has frozen the city. They confront the witch, the protagonist faints and the witch escapes. After the protagonist wakes in a scholar's cabin, they discuss the Royal Library. The scholar believes an old account of defeating the witch, stored at the Royal Library, may help them, but the way is blocked by monsters. The group and the scholar agree to fight the monsters to travel to the Royal Library. An example of an improvement seeker is the protagonist in STVA. They unearth an artefact on their farm. A quest appears in their journal saying that Gunther, the curator at the Museum (also a library), can decipher the properties of the artefact. A town resident also discusses the usefulness of the library. The player can choose to visit the library to talk with Gunther.

*4.1.4 What did users do in libraries?* Problem solvers sought out a relevant piece of information, artefact/object, or person in libraries to achieve their goals. This could be easily accessible or indicated (such as a glowing book). Or it could require exploring many spaces, using tools (such as card catalogues) and overcoming barriers (such as solving a riddle). These provided historical information or presented helpful characters that addressed short-term problems. For example, the DQ11 protagonist finds solutions from the past, and the ACOR protagonist meets a hidden ally.

Improvement seekers brought artefacts and objects to the library, talked with librarians and other users, and used the library spaces and services to study. In each game, companions sat in the libraries and could discuss their personal lives with the protagonists. In DAIN and STVA, librarians and researchers analysed artefacts and provided rewards or bonuses to the protagonists. In PSA5, the protagonist improved their knowledge and other abilities by studying at a library.

In every library, the protagonists could communicate with others, either with other users (ACOR, DAIN, DQ11, FRAB), librarians (EASH, MIS2, TSIC) or both (the other games in the study sample). In DAIN, DQ11, FFA8, NIER, NITW, PSA5 and SKIM they could explore the libraries with their companions, either through bantering or discussing specific information and barriers they encounter. These companions, the secondary characters, influenced the protagonists and the overall narratives. Other users or librarians were generally tertiary characters, in that they affected the protagonists only in a trivial manner. They remained in stationary positions and shared short, repetitive dialogue or conversations. Exceptions included the prominent librarians in NIER, ONSH and PSY2. These librarians helped or hindered the protagonists in pivotal moments or appeared in multiple locations. Generally, the companions and other users were characterised as more intellectual or bookish than the protagonists. They could be scholars (ACOR), enthusiastic readers (ONSH) or curious patrons (NIER). However, in DQ11, FFA8 and STVA, unenthusiastic young people were depicted as being intimidated or confused by the libraries.

In eight of the games, barriers within the libraries hindered the protagonists' progress. DQ11, NIER and PSY2 depicted enemies the protagonist could defeat. In PSY2 the librarian fought the protagonist. EASH, FRAB, NIER, ONSH and PSA5 depicted restricted areas (such

as a place to read) within the library. These were either unlocked as time passed (NIER and PSA5), required the protagonists to solve a puzzle (FRAB) or to obtain permission from elsewhere (EASH and ONSH) to access the area. In SKIM for instance, the protagonist left the library to find and return stolen books for the librarian to gain access. While not hindering progress, the protagonists could express intimidation or enthusiasm at the size of the libraries, such as in ACOR when Bayek expressed concerns about finding his companion in such a large library.

Libraries provided various artefacts/objects, people and environments the protagonists could serendipitously observe and interact with. Background information about the world immersed the protagonists in its wider societal and historical contexts (for example, past tragedies). The protagonists could also bond with their companions while discussing these contexts. In ACOR, DAIN, NIER, NITW, SKIM and TSIC, books or conversations presented side opportunities and quests the protagonists could pursue beyond their main objectives (such as getting directions to a hidden treasure). These immersive activities formed a deeper layer of narrative around simple information needs and seeking. For example, in NITW, Mae and her friend Bea look for clues about historical ghost activity in the local library. They ask the librarian where to find the microfiche reader. Bea, more comfortable in libraries, teaches Mae to use the reader. Mae reads some newspaper clippings and finds three places where ghosts had been spotted. She also reads clippings about past labour strikes and a symbolic tooth (which the player can later identify and gift to someone). Mae and Bea bond and discuss their town, its history and their memories. Before leaving the library, they witness together a poetry reading about the struggles of living in their town.

*4.1.5 How did users use information?* Problem solvers used information from libraries as clues or solutions to overcome short-term barriers. For example, in MIS2, Guybrush used a library card as identification to buy alcohol. Improvement seekers used their improved abilities and better relationships to help their journeys more generally. For example, in DAIN the protagonist learned to fighting enemies more effectively. Information from libraries helped both types of protagonists to understand their worlds and contexts as they progressed.

Problem solvers often struggled to immediately understand information or had to bring it to another character who directed them. Librarians helped them understand information or problems with their knowledge of history, old artefacts and local issues (EASH, NIER, ONSH, SKIM and STVA). Other users, companions or mentors helped the problem solvers in similar ways (ACOR, DQ11, NIER, NITW and PSY2). This equipped the protagonists to use the information they found in libraries. In addition to their main objectives the protagonists could pursue less significant side opportunities. This could be clearly communicated to the player, such as when a conversation initiated a side quest to help that character. Or the player would have to recognise the opportunity, such as in NITW where they can read an article about an artefact and later gift it to another character. Problem solvers were depicted as rarely returned to the libraries unless they encountered barriers within them. However, if they returned, they could encounter changes in dialogue or the library environments that reflected their previous actions (DQ11, EASH, NIER, PSY2, SKIM and TSIC). For example, in PSY2 Raz encounters a library and a domineering librarian in Cassie's "Mental World" and helps Cassie to overcome the librarian and resolve her psychological issues. Cassie then helps Raz to remove an obstacle. After this, Raz can return to Cassie's Mental World and talk with the now friendly librarian. He can also use abilities he learned in the library to reach previously inaccessible areas.

*4.1.6 How did the games end?* The narrative of the games that ended did so with the protagonists succeeding in their wider objectives. However, their success could be subverted or complicated in games such as NIER and TSIC (where they encountered an unforeseen danger in their desires). ACOR, DQ11, NIER, ONSH, PSY2 and SKIM allowed players to continue exploring after the narrative ended. STVA eschewed an ending altogether. Though the libraries were useful, they rarely played a large part in the climax of the narratives. Nevertheless, in FFA8

and ONSH the librarians appeared briefly in credit sequences. In NIER and PSY2, librarians hindered and helped the protagonist respectively. Popola, the librarian in NIER, revealed that she had manipulated the protagonist throughout the game. She explained that the protagonist and her were androids. She wanted to assist the repopulation of human life (that died out centuries ago). The protagonist realised the spirits he fought to save his sister were in fact human. Despite this surprise, he defeated Popola and saved his sister.

#### 4.2 Textual analysis

Carr's (2019) lenses were applied iteratively to any element related to library use and the protagonists' associated information behaviour in the games. The lenses helped develop a deeper analysis of the games' meanings across multiple levels inherent to video games, such as mechanics and narrative.

The "outsider" status of the protagonists in their worlds caused a generally lonely feeling for a player. For example, in TSIC, the researcher-player constantly had to review a map of the town to navigate its jumbled streets as the residents were reluctant to help an outsider. However, all the games thematically and mechanically encouraged building friendships or relationships. In FFA8 the narrative followed the protagonist journey to learn to trust others. PSA5 provided bonuses to players if they improved their relationships.

Libraries in the games reinforced these themes and feelings. Librarians could be hostile, books unreadable or other users uncommunicative. The "insiders" remained stationary while the protagonists came and went. On the other hand, the protagonists shared information, learned about their worlds and bonded with others. This led to touching scenes such as the poetry reading in NITW, which connected the protagonist to the emotions of the townspeople. In a similar instance, the protagonist of EASH, a traveller, encounters mysterious drummers in the forest and visits the library to learn more about them. The librarian mentions that a patron has asked her to direct people interested in drumming to a book. A note in the book invites the protagonist to meet a drummer elsewhere. The drummer, Alejo, is one of the local, reclusive indigenous peoples. The protagonist visits his home, where they reassure Alejo about the "outsiders", the non-indigenous peoples inhabiting the island. The protagonist leaves as a "Friend of the First Folk."

The library settings in the games (except for DQ11 and PSY2) are often depicted as safe intermissions or breaks between actions and conflicts. They could appear uncomfortable or eerie, such as in PSA5, where other library users whispered rumours about the protagonist. In other cases, the music and aural ambience of the libraries, often peaceful, cultivated a feeling of safety (EASH, FFA8, FRAB, NIER, ONSH and STVA). These conditions encouraged calm exploration of the library spaces. For example, in FRAB, outside the library, Fran sees disturbing hallucinations. In contrast, the library is a quiet hollow inside a tree-like, fantastical castle. Candles burn and pages turn softly. Fran meets a stropky user, but she pleasantly converses with a couple of others. She finds a puppet-version of a scary-looking figure she has previously met. She decides that if the figure appears in the library, it may be nicer than she has assumed.

#### 4.3 Narrative context

During the analysis, data were contextualised in view of the associated genre, settings and subjects of the various video games. All the games were single player, and so emphasised the individual journeys of protagonists connecting with the worlds and people around them. Adventure games encouraged problem-solving and observation but limited the protagonist's movement. Roleplaying games encouraged bonding with characters, everyday life experiences and improvement seeking but left much of this to the player's preferences. Action games encouraged movement and combat but simplified social and intellectual interactions. These genre conventions affected the protagonists' library use by setting the boundaries around their abilities and concerns.

Books and familiar library tools (appropriate to their time) appeared in all the games regardless of their fantastical or futuristic settings. In FFA8 the protagonist said they preferred computers as they were accordingly more efficient than the library. While in other games, such as DQ11 and NITW, libraries were necessary resources for learning about the past or contemporary issues. This occurred in decaying or scarce worlds, particularly in the peri and post-apocalyptic genres. These settings, such as NIER and ONSH, emphasised the inaccessibility of the past outside of library spaces, depicting libraries as important places and resources for past records and historical documentation.

The subjects of the games interweaved with the themes and events that occurred in the libraries. All the games featured young protagonists and depict them connecting with others to learn about themselves and the worlds around them. The libraries in these games contributed to this desire to grow and learn, by providing useful advice and information and immersing them in the wider issues and contexts of their worlds (both past and present).

## 5. Discussion

### 5.1 *Depiction of library use*

Library use in video games supports protagonists' achievement of their goals and enriches the narrative settings of the games. Problem solvers visit libraries when they encounter obstacles. Solutions to the obstacles include finding clues, helpful users or solutions/methods from the past. Improvement seekers study, contribute materials for research or talk with companions. Both [Holmström \(2019\)](#) and [Kussler \(2020\)](#) also observe such depiction of short-term and incremental benefits of libraries in video games. One observation of the current study is that in NIER and PSY2, library use holds more thematic and narrative weight because it involves interactions or conflicts with important characters such as the librarians. Another considerable aspect of library use observed in all the games is that serendipitous library use leads to the protagonists understanding their worlds and contexts in the long-term. This aligns with [Pierce's \(2004\)](#) suggestion that fictional young people develop their identities and connect to their wider communities in libraries. As Kussler observes, this immersion also helps game players make decisions within a fictional context. It is observed in the current study that library use in video games interweaves functional and thematic elements to help the users for both short-term and long-term benefits.

Libraries help users solve problems, improve themselves, and bond with other characters in both troubled and peaceful worlds. Ten of the fifteen games depict conflict and unrest preceding, coinciding or following a catastrophic event. In seven of these games the protagonist must visit libraries (DQ11, NIER, NITW, ONSH, PSY2, SKIM and TSIC). [Duxfield and Liew \(2022\)](#) connect the prominence of libraries in fictional depictions of catastrophes with the value of knowledge in times of scarcity. In the current study, it is found that even in peaceful times, such as the one depicted in EASH, the library is necessary in that the player must interact with it to access an area where they can complete their main quest. The mechanical genre of games, in addition to thematic genre, account for the function and importance of the libraries. Roleplaying games emphasise optional improvement seeking, as opposed to the problem solving of action and adventure games. This aligns with [Holmström's \(2019\)](#) argument that video game genres influence the depictions of libraries. However, consistently across genres, as [Kussler \(2020\)](#) and [Duxfield and Liew \(2022\)](#) observe, physical books and library tools contribute to library use. While these studies are focused on medieval and science fiction media, it applies also for the historical, contemporary and futuristic settings depicted in the current sample of video games.

Users mostly find safe places in libraries. However, libraries are not always immediately welcoming. In the libraries, protagonists can experience uncomfortable interactions, intimidating spaces or barriers to progress. This aligns with previous discourses around fictional libraries as scary and rigid places ([Radford and Radford, 2001](#); [Tewell, 2014](#)). It is

observed nevertheless that in the video games, the protagonists eventually overcome barriers presented to them and achieve their goals. This is a common feature of video games (Holmström, 2019). In the context of processes, libraries also act as places to rest and solve problems between conflicts.

In summary, in the video games studied, while libraries may only be a small part of the users' personal processes, encountering and using the library allow them to connect with or immerse themselves in the worlds they explore. Libraries are generally depicted as safe and helpful places in both hostile and peaceful settings. The focus of this study on the library users, rather than librarians, results in a more positive impression of their experiences, in the vein of Duxfield and Liew (2022). The games in this sample emphasise the role of libraries in connecting users to their wider communities and social contexts.

### *5.2 Depiction of information behaviour around library use*

Information seeking and use among most users in libraries depicted in the video games is simple and streamlined rather than circuitous and iterative. They arrive knowing that they need a specific thing or to fulfil a specific information need and they eventually succeed. The exceptions are improvement seekers, whose seeking depends on player preference (rather than fictional needs), and more elaborate depictions, such as in PSY2 where Raz encounters difficulties in the library. The general trend among users in video games differs to Hicks and Whippley's (2013) observation of piecemeal information seeking in the television show "Buffy". It also contrasts with previous observation of multifaceted and recursive human information behaviour (Case *et al.*, 2016). This could be because of the narrative context of the depictions. The games provide explicit objectives for players, who progress from one to the next in a linear fashion or choose what to pursue.

Information sharing is common inside and outside of libraries. In seven games, other characters direct protagonists to libraries (ACOR, DAIN, FRAB, NIER, NITW, PSA5 and STVA). The protagonists can visit libraries with companions or discuss information they find with librarians and other users. They then seek guidance outside the libraries from experts or those who can direct them. Libraries appear as conversational and collaborative places, similar to Estill's (2007) observation of the library in "Buffy". This helps the protagonists because they need to understand information and situations through locals, experts and friends. This understanding helps them contextualise and act on the information they find in libraries to solve short-term problems or improve themselves and their relationships.

The games depict insider and outsider dynamics that intertwine with library use and demonstrate the complexity of these dynamics. In thirteen of the fifteen games, the protagonists arrive in their worlds as outsiders (such as travellers or criminals). In five games, libraries require community permission to enter, but once the protagonists enter the libraries, they can access the shared knowledge, which Chatman (1996) defines as "insider" behaviour. In FRAB, MIS2, PSA5, PSY2 and SKIM, librarians and other users intimidate or block the protagonists initially, and then eventually help, or the protagonists outwit them. In other games, there are eager helpers, and library users are eager to find and share information. This conflicts with Chatman's conception of marginalised people avoiding interaction outside their groups. The libraries also complicate the state of being an insider. For example, in NIER the protagonist discovers a librarian's vast historical conspiracy. In NITW the protagonist returns home and begins to confront her town's history in the library. These differences from Chatman's conceptions could arise from the individualistic narratives of the games (a marginalised person as opposed to community). However, they align with Floegel and Costello's (2021) arguments for the fluidity of insider and outsider dynamics, and the richness of marginalised information worlds.

In summary, video games depict information behaviour in libraries as simpler than real-life behaviour or depictions in other media. This is likely because of their audiences and



genre conventions. Fictional library users in video games accept direction from others and share information within and outside libraries to understand how to use it. This is important for the users as outsiders in their worlds, but even insiders confront surprising knowledge in libraries. Insider and outsider dynamics are fluid and contextual, illustrating a gradual shift in literature on information poverty (Chatman, 1996). This highlights a key aspect of depictions of library use in video games, in that information seeking and use is highly embedded in the fictional users' social contexts and relationships. While it may lead to real-life users underestimating the complexities of library use, it emphasises the role of libraries in helping both insiders and outsiders access information.

### *5.3 Depiction of character traits*

The protagonists in some games possess traits and abilities that help them solve problems librarians and other users cannot solve in segments involving libraries. The protagonists' personalities and abilities allow them to explore, traverse, communicate and connect various elements. Librarians and other library users are generally characterised as intellectual or bookish and are often stationary, there to help contextualise or decipher information with their local or specialised knowledge. Protagonists help others find or order information, while others help them understand it and enable them to act. This contrasts with Radford and Radford's (2001) conception of library users as chaotic and rebellious. In the games in this study sample, the protagonists improve or help libraries more than they threaten them. Adventurous protagonists and more intellectual librarians and users help each other. This dichotomy observed between active protagonists and intellectual others is consistent with Estill's (2007) observation that Giles, the librarian in "Buffy", is the "mind" and Buffy, the protagonist, is the "body". Brock's (2011) discussion on video games mechanics contextualises the prominence of this dichotomy in games. Brock argues that games emphasise how players can fight and exploit other characters over how they can think and communicate with them. In games, protagonists need help exploring but they use information in ways others cannot. They can make use of acquired information and knowledge to defeat enemies, solve puzzles and explore inhospitable places.

Most of the protagonists appear as outsiders learning the ways of unfamiliar worlds. They are primarily young, male, sexually heterosexual or ambiguous, and Caucasian or Japanese. Other users in the libraries are more demographically diverse but reflect their community and setting (such as an academy or a fantastical culture). Brock (2011) criticises this common narrative in which straight white males heroically adventure through exotic lands. Even in games when a player can create their own character, a white male is often the default choice. This again complicates insider and outsider dynamics. The games rarely depict protagonists as belonging to real-life marginalised communities (except their status as outlaws or unemployed people) but present them as outsiders in strange worlds. Regardless of the characters' demographics, users are mostly depicted as outsiders. Library anxiety is a common concept applied to real-life library users that describes the feeling of being less familiar with libraries than other users (Tewell, 2014). This concept parallels how the fictional users approach libraries as unfamiliar or confronting spaces. However, through interactions with the library in all the games, the protagonists overcome these anxieties. This is encouraging as it suggests that real-life library users can similarly overcome their library anxiety through further interactions with appropriate support offered by the libraries.

In summary, video games generally depict library users as adventurous protagonists and their companions, or intellectual and bookish others. This aligns with previous literature on the "body" and "mind" dichotomy but contrasts with users as chaotic forces. Demographically the protagonists are less diverse than other users and librarians. Depicting the protagonists as outsiders reflects common conventions of video games, but

it also suggests that real-life users may feel unfamiliar or outside of libraries. The games depict users overcoming difficulties. The current study reveals a more optimistic depiction of library use than those documented in [Radford and Radford \(2001\)](#) or [Tewell \(2014\)](#). It suggests that users and libraries can overcome library anxiety.

#### *5.4 Assumptions and limitations*

The small sample size of 15 video games is acknowledged, as well as the subjective nature of individual coding and interpretation. The insights gained are not meant to be generalisable and the value of the research is based on the supposition that depiction of users and use of libraries in video games can provide insightful indications into real-life perceptions of the value and function of libraries. There are avenues for further research. Future research can involve a larger sample of video games from different genres and undertake a comparative study across the sample. Researchers could also consider the cultural differences of games originated in different countries/cultures and compare their characterisation and depiction of library users and library use in different sociocultural contexts.

### **6. Conclusion**

This research broadens understanding of depictions of library use in popular culture. It contributes new insights from a previously overlooked medium in similar scholarly investigations, namely video games. It is observed that fictional users in video games make use of libraries to not only solve short-term problems but learn and gain knowledge that can be useful in the longer term, as well as to interact and bond with other individuals (other characters in the games). While being a small part of their overall processes, the library users also use the learning and knowledge they have gained from using the libraries to immerse themselves in the histories and concerns of the peoples of unfamiliar worlds. Libraries are set up as safe places for users to immerse themselves in the wider contexts of their worlds, presenting more positive images of library use than previous works have documented in films, television and literature. It encourages already established roles for libraries: providing information, places to study and gather, and helping users to familiarise with and immerse in their wider communities, and the associated cultural and historical contexts.

Fictional library users in the study sample are presented as active and adventurous but also as outsiders to the library, so they often rely on others to understand newly encountered information. These insights can be valuable when designing and planning real library spaces and services, and how these encourage or discourage these activities (such as allowance and locations for library spaces where conversations are encouraged). Fictional libraries immerse users, both outsiders and insiders, in their worlds. Real libraries could find ways to serendipitously engage users while fulfilling their core tasks, such as through strategic displays of local history in the library spaces.

Fictional users are generally characterised as being unfamiliar with libraries at first, and the depictions of information behaviour are often simple. However, with help and guidance, they start making more meaningful use of the libraries in the game world. A takeaway point from this for libraries is that they could consider introducing appropriate onboarding support services for users new to their library or new to a specific resource or service offered by the library, as well as for “outsider” users such as tourists, if these are not already existing. Depicting player-controlled characters as requiring help but also being able to help librarians and other users in turn, suggests that real users and non-users may feel they can contribute to libraries. Libraries in the real world should prioritise helping users overcome library anxiety, and to acknowledge that users may have something to offer to libraries, such as volunteering for library programmes or as community partners.

Fictional library users in the sampled games seek and use information in a streamlined fashion. They can navigate spaces, converse or use tools to find relevant information in libraries. Others often direct them to libraries, or they must share information with others to understand and use it. These interpersonal connections situate information seeking and use in individuals' social contexts. The users are often outsiders immersing themselves in their worlds, but libraries also confront insiders with uncomfortable revelations or histories. This fluidity of insider and outsider dynamics reflects critiques of Chatman's (1996) theories. These depictions may reflect or cause simplistic perceptions of library use among real users and non-users. However, they also depict libraries as being useful for both outsiders and insiders.

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(The Appendix follows overleaf)

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# Depiction of library use in video games

| Code                         | Definition                                                                                                                                             | Example                                                                                                                                       | Lens (Carr, 2019)                                                                   | Narrative context                                                                       |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <i>Information behaviour</i> |                                                                                                                                                        |                                                                                                                                               |                                                                                     |                                                                                         |
| Information need             | Users recognising their “knowledge is inadequate to satisfy a goal [they] have.” (Case <i>et al.</i> , 2016, p. 6)                                     | NITW: <i>Mae thinks a ghost is haunting her town, but the police don’t believe her, and she doesn’t know about the town’s ghost history</i>   | Thematic: <i>Mae is unaware of the history of her town despite growing up there</i> | Setting: <i>a decaying (i.e. ghostly) town</i>                                          |
| Information seeking          | Users’ “conscious effort to acquire information in response to a need or gap in [their] knowledge.” (Case <i>et al.</i> , 2016, p. 6)                  | DQ11: <i>the protagonist and his group look for a book in a library to understand how to defeat a witch</i>                                   | Structural: <i>the player cannot find or beat the witch till they read the book</i> |                                                                                         |
| Information use              | What users do “with the information acquired through seeking, serendipity, or other means.” (Case <i>et al.</i> , 2016, p. 6)                          | MIS2: <i>Guybrush uses a library card to buy grog at a bar so he can win a spitting competition so he can charter a boat to find treasure</i> |                                                                                     |                                                                                         |
| Information sharing          | When users share information with others in work or social contexts. It’s collaborative if they share a goal. (Case <i>et al.</i> , 2016, pp. 374–375) | FRAB: <i>Fran brings the wizard’s book to him, and he tells her where to find the next clue</i>                                               |                                                                                     | Subject: <i>a courageous girl escaping terrible conditions with the help of friends</i> |
| Unintentional behaviours     | Users’ “serendipitous actions (such as glimpsing or encountering information).” (Case <i>et al.</i> , 2016, p. 370)                                    | PSY2: <i>Raz encounters The Librarian, who forces him to find two books for her</i>                                                           |                                                                                     |                                                                                         |
| Internal motivations         | “Thoughts and emotions that accompany” users’ information behaviour. (Case <i>et al.</i> , 2016, p. 370)                                               | ONSH: <i>Niko hurries to the library because “people are counting on us!” to save the world</i>                                               |                                                                                     |                                                                                         |
| External motivations         | The “influence of contextual elements like situation, time, affect, culture, and geography” on users. (Case <i>et al.</i> , 2016, p. 370)              | SKIM: <i>the head of the academy tells the protagonist to ask the librarian for books</i>                                                     |                                                                                     |                                                                                         |
| <i>Library use</i>           |                                                                                                                                                        |                                                                                                                                               |                                                                                     |                                                                                         |
| Process                      | “How an individual uses a library or information to achieve an end.” (Fleming-May, 2011, p. 309)                                                       | ACOR: <i>Bayek wants to find Aya in the library, who will help him kill an evil cultist</i>                                                   |                                                                                     |                                                                                         |
| Task                         | The actions of the library user to complete their process                                                                                              | EASH: <i>the protagonist asks the librarian for access to a restricted area</i>                                                               |                                                                                     |                                                                                         |
| Outcome                      | Whether the library user is successful or unsuccessful in completing their process                                                                     | PSA5: <i>the protagonist improves their “Knowledge”, but it’s lessened because they’re distracted by noise in the library</i>                 |                                                                                     | Genre: <i>the game, an RPG, is based around improving “Stats” through activities</i>    |

(continued)

**Table A1.**  
Coding scheme

|  | Code                        | Definition                                                                                                 | Example                                                                                                                                    | Lens (Carr, 2019)                                                                         | Narrative context |
|--|-----------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------|
|  | Social immersion            | A character learning about their world/social context                                                      | TSIC: <i>Charles is shocked to meet the librarian, whose mouth is sewn shut in a local form of punishment</i>                              |                                                                                           |                   |
|  | <i>Character Traits</i>     |                                                                                                            |                                                                                                                                            |                                                                                           |                   |
|  | Importance                  | Primary (player), secondary (impacts player), tertiary (background). (Carroll, 2021)                       | STVA: <i>the protagonist (primary) can build relationships with other library users (secondary)</i>                                        |                                                                                           |                   |
|  | Appearance                  | Users' visual signifiers and physical characteristics. (Carroll, 2021)                                     | NIER: <i>the protagonist wears hunting gear and carries weapons</i>                                                                        |                                                                                           |                   |
|  | Traits                      | Significant personality traits or features                                                                 | FFA8: <i>Squall is brash and sullen, not open to others</i>                                                                                |                                                                                           |                   |
|  | Demographic characteristics | "Species, race/ethnicity, gender, age, job, and sexual orientation." (Carroll, 2021, p. 24)                | DAIN: <i>the player can create a character, but a Caucasian male is the first option</i>                                                   | Intertextual: <i>games have been critiqued for often presenting white males as heroes</i> |                   |
|  | Mechanical characteristics  | What can the player do? How can they interact with libraries? How can non-player users interact with them? | FFA8: <i>other users can battle Squall with trading cards, but the player must initiate it</i>                                             |                                                                                           |                   |
|  | Outsiders                   | The character is somehow outside of communities/societies in the game world or marginalised                | ACOR: <i>Bayek is a traditional Medjay, a defender of the pharaoh. He sees Romans as outsiders in Egypt, but they see him as an outlaw</i> |                                                                                           |                   |

Table A1.

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